

Cross-Feature Attention for Architecture-Level CPU Power Modeling: An FT-Transformer Approach on the ArchPower Dataset

Jingbo Jiang

Abstract

Accurate architecture-level power estimation is critical for early-stage CPU design-space exploration. We propose using the Feature Tokenizer Transformer (FT-Transformer) for component-level CPU power prediction. FT-Transformer tokenizes each of 81 event features into a learned embedding and applies multi-head self-attention to model cross-feature interactions—capturing how coupled microarchitectural activities jointly determine power. On the ArchPower dataset (200 samples, 2 RISC-V architectures), FT-Transformer achieves 3.61%/5.33% MAPE (BOOM/XiangShan), outperforming Ridge (4.36%/5.95%) and MLP (6.29%/7.76%) baselines. While XGBoost currently edges ahead (2.36%/4.26%) due to its small-data advantage, we propose collecting an expanded dataset for the final report where we expect FT-Transformer to achieve the best accuracy.

1. Introduction

Power consumption is the primary design constraint for modern processors. Accurate power evaluation traditionally requires the full RTL implementation, logic synthesis, and signoff-grade power analysis—a process that easily takes months and precludes rapid design-space exploration at the architecture level [6]. Classical analytical tools such as McPAT [2] attempt to fill this gap but are known to exhibit large modeling errors on modern out-of-order CPUs.

Recent work has proposed ML-calibrated analytical models. McPAT-Calib [3] trains XGBoost regressors to correct McPAT’s component-level estimates, while PANDA [5] introduces physics-aware resource functions that encode hardware-dependent scaling prior to ML training. Neural-network-based approaches have also been explored [4], but existing methods treat input features independently—missing the rich cross-feature interactions inherent in CPU power behavior. For example, branch misprediction rates interact with instruction fetch bandwidth to determine front-

end power, and cache miss rates couple with memory controller traffic to drive back-end power. Models that capture these pairwise interactions should yield both better predictions and more interpretable insights.

We propose applying the Feature Tokenizer Transformer (FT-Transformer) [1] to architecture-level power modeling. FT-Transformer tokenizes each scalar feature into a learned embedding, then applies multi-head self-attention over all feature tokens. This design is uniquely suited to power modeling because: (1) each attention head learns a distinct interaction pattern (e.g., pipeline-stage coupling vs. memory-hierarchy coupling); (2) the 81×81 attention matrices are directly interpretable as event-to-event interaction maps; and (3) multi-target regression provides implicit regularization critical for the small-data regime.

Our contributions at this milestone are: (1) an event-only preprocessing pipeline that factors out hardware dependence via PANDA’s resource functions (101→81 features); (2) demonstrating that FT-Transformer outperforms both Ridge and MLP baselines, validating cross-feature attention for power prediction; and (3) a data scaling hypothesis arguing that XGBoost’s current lead is an artifact of the limited dataset size.

2. Problem Statement

We use the ArchPower dataset [6], an open-source benchmark with 200 CPU data samples from two RISC-V architectures: BOOM (15 configs $\times$ 8 workloads = 120 samples) and XiangShan (10 configs $\times$ 8 workloads = 80 samples). Each sample has a 101-d feature vector (20 hardware parameters + 81 event parameters) and power labels for 12 components $\times$ 5 power groups.

Task. Given only the 81 event features (after removing hardware dependence), predict the 12-component power vector for unseen CPU configurations. Splits are at the configuration level—all 8 workloads from a config are entirely in train or test—to evaluate cross-configuration generalization.

Evaluation. MAPE, R^2 , and RMSE on a config-level

holdout (20% of configs).

Challenge. The extreme sample-to-feature ratio (120 or 80 samples for 81 features) makes overfitting the dominant risk.

3. Technical Approach

3.1. Data Preprocessing: Event-Only Dataset

Following PANDA [5], we model each component’s power as $P_c = R_c(\mathbf{x}_{\text{hw}}) \cdot f_c(\mathbf{x}_{\text{event}})$ and divide out the hardware-dependent resource function R_c to obtain normalized labels $P'_c = P_c/R_c$. Resource functions are multiplicative for most components (BP, ICache, IFU, RNU, LSU, DCache, ROB), additive for Regfile, lookup-table for ISU, and identity for FU-Pool. After normalization, the 20 hardware-constant columns are removed, yielding an 81-d event-only feature matrix used by all downstream models.

3.2. Proposed Model: FT-Transformer

The core of our approach is the Feature Tokenizer Transformer (FT-Transformer) [1], designed specifically for tabular data. Unlike MLPs that process the concatenated feature vector through dense layers, FT-Transformer tokenizes each feature independently: every scalar event feature x_j is projected into a d -dimensional embedding $\mathbf{t}_j = \mathbf{w}_j x_j + \mathbf{b}_j$. A [CLS] token is appended, and the 82 tokens are processed by transformer blocks with multi-head self-attention.

Why attention captures cross-feature interactions. Power consumption arises from complex interactions between microarchitectural activities—e.g., instruction cache misses increase fetch stalls, reducing issue-queue utilization and functional-unit power. Linear models ignore these interactions; MLPs learn them only implicitly. In contrast, each self-attention head computes an explicit 81×81 interaction matrix $\mathbf{A} = \text{softmax}(\mathbf{QK}^T/\sqrt{d_k})$, where A_{ij} quantifies how strongly event i attends to event j . Multiple heads capture distinct patterns simultaneously—pipeline-stage coupling, memory-hierarchy dependencies, etc.

Architecture and training. We use $d_{\text{token}}=32$ (BOOM) or 48 (XS), 2 transformer blocks, 4 attention heads, and FFN dimension 64 (22K–40K parameters). Training has two phases: (1) self-supervised pre-training via masked feature modeling, where random features are masked and reconstructed to learn feature correlations; (2) supervised fine-tuning with MSE loss on all 12 power components and early stopping. After training, we extract attention weight matrices to construct event-to-event interaction maps.

Table 1. Total-power prediction on BOOM and XS (config-level holdout, val_ratio=0.2). Best deep learning result underlined; best overall in bold.

Model	BOOM		XS	
	MAPE%	R^2	MAPE%	R^2
Ridge	4.36	0.856	5.95	0.905
MLP	6.29	0.629	7.76	0.858
FT-Trans.	<u>3.61</u>	<u>0.925</u>	<u>5.33</u>	<u>0.932</u>
XGBoost	2.36	0.938	4.26	0.945

3.3. Baseline Models

Ridge Regression. Multi-output Ridge (81→12, 984 params)—captures only first-order relationships.

3-Layer MLP. Hidden layers (128, 64, 32) with ReLU and L2 regularization (21K params)—learns nonlinear mappings but models interactions only implicitly.

XGBoost GBDT. Twelve independent regressors (100 trees, depth 6)—excels on small tabular data but lacks explicit cross-feature interaction modeling and interpretability.

4. Intermediate/Preliminary Results

4.1. FT-Transformer Outperforms Ridge and MLP

Table 1 summarizes total-power prediction accuracy. Our FT-Transformer substantially outperforms both baselines: it reduces MAPE by 17% relative to Ridge (3.61% vs. 4.36% on BOOM) and by 43% relative to the MLP (3.61% vs. 6.29%). FT-Transformer achieves R^2 of 0.925 (BOOM) and 0.932 (XS), compared to 0.856/0.905 for Ridge and 0.629/0.858 for the MLP. These gains validate that explicit cross-feature attention improves power prediction over models that either ignore interactions (Ridge) or learn them only implicitly (MLP).

4.2. Component-Level Analysis

Table 2 compares per-component MAPE on BOOM. While XGBoost leads on most components, FT-Transformer outperforms XGBoost on three components: ROB (18.25% vs. 25.75%), RNU (28.71% vs. 39.18%), and Others (30.51% vs. 42.94%). These components’ power depends on complex inter-unit interactions (e.g., ROB occupancy couples with rename width and issue-queue pressure), suggesting that attention provides a genuine advantage where cross-feature interactions dominate.

Table 2. Selected per-component MAPE (%) on BOOM. Components where FT-Transformer outperforms XGBoost in bold.

Component	XGBoost	FT-Trans.
Total	2.36	3.61
BP	1.05	1.04
Regfile	13.24	14.34
ROB	25.75	18.25
RNU	39.18	28.71
Others	42.94	30.51

4.3. Why XGBoost Leads—and Why This Will Change

XGBoost currently achieves the best total-power MAPE (2.36% / 4.26%), outperforming FT-Transformer by ~ 1.3 percentage points. We attribute this to a well-known phenomenon: tree-based ensembles dominate neural models in the small-data tabular regime [1]. With only 200 samples, gradient-boosted trees benefit from built-in feature selection and sample-efficient splitting. The original FT-Transformer paper [1] shows that this gap narrows and reverses as dataset size grows, motivating our plan to collect a larger dataset for the final report.

4.4. Next Steps

For the final report, we plan to: (1) Collect an expanded dataset with additional CPU configurations and workloads (target 1,000+ samples), testing whether FT-Transformer surpasses XGBoost as expected; (2) Analyze attention maps to identify previously unknown event interactions for CPU designers; and (3) Refine the model with deeper variants and architectures that enforce the component-to-total summation constraint.

References

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